CSCE 763: Digital Image Processing
Homework #1

Instructor: Spring 2026
University of South Carolina
Solutions by: JC Vaught

Due: 1:15 pm EST, Tuesday, Feb 3

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Problem 1: Smallest Discernible Dot (25 pts)

Thinking purely in geometric terms, estimate the diameter of the smallest printed dot that the eye can discern if the page on which the dot is printed is 0.5 m away from the eyes.
Assumptions:

1. The distance between the center of the lens and the retina along the visual axis is 14 mm.
2. The visual system ceases to detect the dot when the image of the dot on the fovea becomes smaller than the diameter of one receptor (cone) in that area of the retina.
3. The fovea can be modeled as a square array of dimensions 1.5 mm × 1.5 mm, and the cones (about 337,000 in total) and spaces between the cones are distributed uniformly throughout this array.

Step 1: Determine the Diameter of a Single Cone

The fovea is a 1.5 mm × 1.5 mm square with 337,000 cones uniformly distributed. Modeling as a square grid, the number of cones per side is:

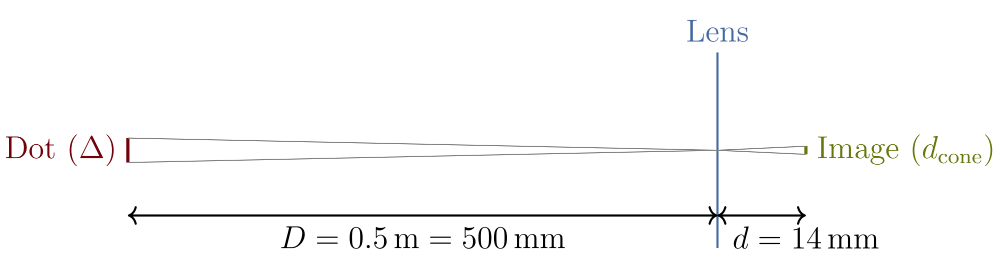
$$n = \sqrt{337{,}000} \approx 580.5$$

The center-to-center spacing (cone diameter including gaps) is:

$$d_{\text{cone}} = \frac{1.5\,\text{mm}}{580.5} \approx 0.00258\,\text{mm} = 2.58\,\mu\text{m}$$

Step 2: Apply Similar Triangles

The dot on the page and its image on the retina are related by similar triangles through the lens center:



By similar triangles:

$$\frac{\Delta}{D} = \frac{d_{\text{cone}}}{d}$$

Step 3: Solve for Minimum Dot Diameter

The smallest detectable image equals one cone diameter: $d_{\text{image}} = d_{\text{cone}} \approx 0.00258\,\text{mm}$.

$$\Delta = D \cdot \frac{d_{\text{cone}}}{d} = 500 \cdot \frac{0.00258}{14} = \frac{1.29}{14} \approx 0.0922\,\text{mm}$$

Results

The diameter of the smallest printed dot that the eye can discern at a distance of $0.5\,\text{m}$ is:

$$\boxed{\Delta \approx 0.0922\,\text{mm} \approx 92.2\,\mu\text{m}}$$

Problem 2: Adjacency of Image Subsets (25 pts)

Consider the two image subsets S_1 and S_2 shown below. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent.

$$
\begin{array}{c|cccc|cccc|c}
 & & S_1 & & & & S_2 & & & \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\
1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\
1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1
\end{array}
$$

Step 1: Identify Pixels in V

The subsets share the boundary between columns 4 (rightmost of S_1) and 5 (leftmost of S_2).

S_1 pixels with value 1: $(3,1)$, $(4,1)$, $(3,2)$, $(2,3)$, $(3,3)$ (using (x,y) coordinates from the grid).

S_2 pixels with value 1: $(7,0)$, $(8,0)$, $(6,1)$, $(5,2)$, $(6,2)$.

The closest cross-boundary pair in V is $p = (4,1) \in S_1$ and $q = (5,2) \in S_2$.

Step 2: Test Each Adjacency Type

(a) 4-adjacency: Two subsets are 4-adjacent if $\exists\, p \in S_1, q \in S_2$ with values in V that are 4-neighbors (share an edge). The nearest pair is $(4,1)$ and $(5,2)$, which differ in *both* coordinates — they are *not* 4-neighbors. No other cross-boundary pair of pixels in V is closer.

$\Rightarrow$ **S_1 and S_2 are NOT 4-adjacent.**

(b) 8-adjacency: $(4,1)$ and $(5,2)$ differ by 1 in each coordinate, so they are diagonal (8-)neighbors. Both have value $1 \in V$.

$\Rightarrow$ **S_1 and S_2 ARE 8-adjacent.**

(c) m-adjacency: Two pixels p, q are m-adjacent if they are 8-adjacent *and* $N_4(p) \cap N_4(q) \cap V = \varnothing$ (no shared 4-neighbor has a value in V). The shared 4-neighbors of $(4,1)$ and $(5,2)$ are $(4,2)$ and $(5,1)$:

$$(4,2) = 0 \notin V, \qquad (5,1) = 0 \notin V.$$

Since the intersection is empty, the m-adjacency condition is satisfied.

$\Rightarrow$ **S_1 and S_2 ARE m-adjacent.**

Results

For $V = \{1\}$:

(a) S_1 and S_2 are **not** 4-adjacent.

(b) S_1 and S_2 **are** 8-adjacent (via $(4, 1) \leftrightarrow (5, 2)$).

(c) S_1 and S_2 **are** m-adjacent (via $(4, 1) \leftrightarrow (5, 2)$, since $N_4(p) \cap N_4(q) \cap V = \varnothing$).

Problem 3: Shortest Paths (25 pts)

Consider the image segment shown below.

(a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4-, 8-, and m-path between p and q. If a particular path does not exist between these two points, explain why.

(b) Repeat for $V = \{1, 2\}$.

			(q)
3	1	2	1
2	0	0	2
1	2	1	1
1	0	1	2
(p)			

Problem 3(a): $V = \{0, 1\}$

Step 1: Identify Pixels in V

Using (c, r) (column, row) coordinates with the origin at the top-left:

$$\begin{array}{cccc} & & & q \\ 3 & \mathbf{1} & 2 & \boxed{\mathbf{1}} \\ 2 & \mathbf{0} & \mathbf{0} & 2 \\ \mathbf{1} & 2 & \mathbf{1} & \mathbf{1} \\ \boxed{\mathbf{1}} & \mathbf{0} & \mathbf{1} & 2 \\ p & & & \end{array}$$

Shaded cells have values in V. $p = (0, 3) = 1 \in V$, $q = (3, 0) = 1 \in V$.

Step 2: Shortest 4-Path

For a 4-path, each step moves to a horizontal or vertical neighbor in V.
Pixel $q = (3, 0)$ has 4-neighbors $(2, 0) = 2 \notin V$ and $(3, 1) = 2 \notin V$. Therefore q is **isolated** in V under 4-connectivity.
$\Rightarrow$ **No 4-path exists.**

Step 3: Shortest 8-Path

For an 8-path, each step moves to any of the 8 surrounding neighbors in V.
$q = (3, 0)$ has 8-neighbor $(2, 1) = 0 \in V$, so q is reachable. One shortest path:

$$(0, 3) \to (0, 2) \to (1, 1) \to (2, 1) \to (3, 0)$$

- $(0, 3) \to (0, 2)$: 4-adj, values $1, 1 \in V$ ✓
- $(0, 2) \to (1, 1)$: diagonal, values $1, 0 \in V$ ✓
- $(1, 1) \to (2, 1)$: 4-adj, values $0, 0 \in V$ ✓
- $(2, 1) \to (3, 0)$: diagonal, values $0, 1 \in V$ ✓

Length = **4**. No shorter path exists (minimum diagonal steps ≥ 3, and no 3-step path can be constructed).

Step 4: Shortest m-Path

For m-adjacency, a diagonal step is allowed only if the two shared 4-neighbors are not both in V.
Checking each diagonal in the 8-path above:

- $(0, 2) \to (1, 1)$: shared 4-neighbors are $(0, 1) = 2 \notin V$ and $(1, 2) = 2 \notin V$. ✓
- $(2, 1) \to (3, 0)$: shared 4-neighbors are $(2, 0) = 2 \notin V$ and $(3, 1) = 2 \notin V$. ✓

All diagonal steps satisfy the m-adjacency condition, so the same path is valid.
Length $=$ **4**.

Results

For $V = \{0, 1\}$:

- **Shortest 4-path:** Does not exist (q is 4-isolated in V).
- **Shortest 8-path:** Length **4**: $(0, 3) \to (0, 2) \to (1, 1) \to (2, 1) \to (3, 0)$.
- **Shortest m-path:** Length **4**: same path (all diagonals satisfy m-adjacency).

Problem 3(b): $V = \{1, 2\}$

Step 1: Identify Pixels in V

$$\begin{array}{cccc}
 & & & q \\
3 & \mathbf{1} & \mathbf{2} & \boxed{\mathbf{1}} \\
\mathbf{2} & 0 & 0 & \mathbf{2} \\
\mathbf{1} & \mathbf{2} & \mathbf{1} & \mathbf{1} \\
\boxed{\mathbf{1}} & 0 & \mathbf{1} & \mathbf{2} \\
p & & &
\end{array}$$

Shaded cells have values in V. Both p and q are now well-connected.

Step 2: Shortest 4-Path

A valid 4-path:

$$(0,3) \to (0,2) \to (1,2) \to (2,2) \to (3,2) \to (3,1) \to (3,0)$$

Length = **6**. Every step is a 4-neighbor in V. No shorter 4-path exists because the middle rows have 0-valued pixels at $(1,1)$ and $(2,1)$ that block more direct routes.

Step 3: Shortest 8-Path

Using diagonal moves:

$$(0,3) \to (1,2) \to (2,2) \to (3,1) \to (3,0)$$

- $(0,3) \to (1,2)$: diagonal, values $1, 2 \in V$ ✓
- $(1,2) \to (2,2)$: 4-adj, values $2, 1 \in V$ ✓
- $(2,2) \to (3,1)$: diagonal, values $1, 2 \in V$ ✓
- $(3,1) \to (3,0)$: 4-adj, values $2, 1 \in V$ ✓

Length = **4**. No 3-step path exists.

Step 4: Shortest m-Path

Check each diagonal step:

- $(0,3) \to (1,2)$: shared 4-neighbors are $(0,2) = 1 \in V$ and $(1,3) = 0 \notin V$. Since $(0,2) \in V$, this diagonal is **not** m-adjacent.
- $(2,2) \to (3,1)$: shared 4-neighbors are $(2,1) = 0 \notin V$ and $(3,2) = 1 \in V$. Since $(3,2) \in V$, this diagonal is **not** m-adjacent.

Both diagonals fail. We must replace each with two 4-adjacent steps. The shortest m-path:

$$(0,3) \to (0,2) \to (1,2) \to (2,2) \to (3,2) \to (3,1) \to (3,0)$$

Length = **6**.

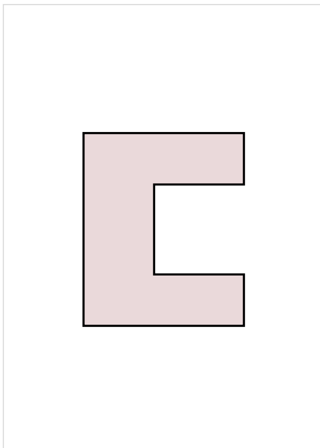
Results

For $V = \{1,2\}$:

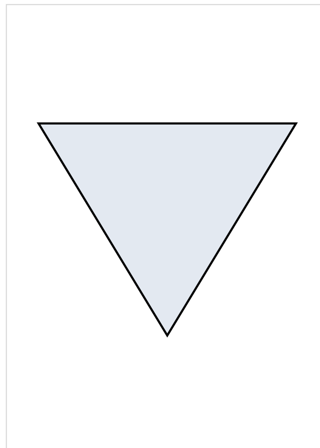
- **Shortest 4-path:** Length **6**: $(0,3) \to (0,2) \to (1,2) \to (2,2) \to (3,2) \to (3,1) \to (3,0)$.
- **Shortest 8-path:** Length **4**: $(0,3) \to (1,2) \to (2,2) \to (3,1) \to (3,0)$.
- **Shortest m-path:** Length **6**: $(0,3) \to (0,2) \to (1,2) \to (2,2) \to (3,2) \to (3,1) \to (3,0)$.

Problem 4: Set Expressions (25 pts)

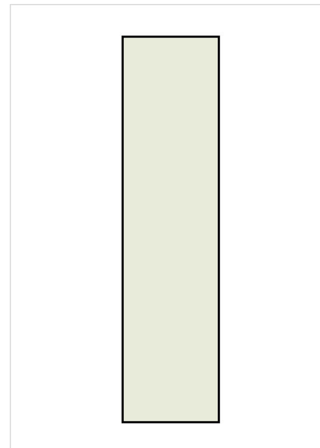
Give expressions for the sets shown shaded in the following figure in terms of sets A, B, and C. The shaded areas in each figure constitute one set, so give one expression for each of the three figures.



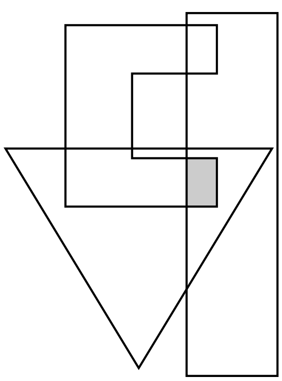
A



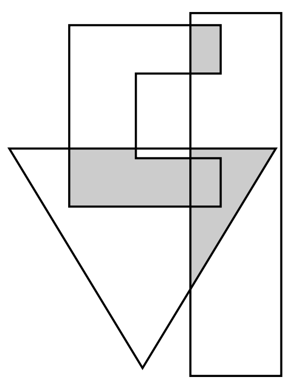
B



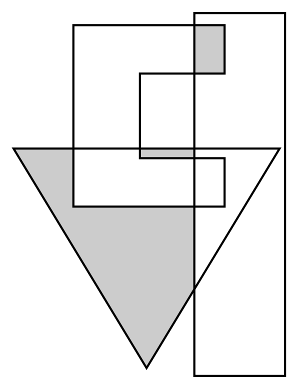
C



(i)



(ii)



(iii)

Step 1: Analyze Each Shaded Region

We identify the shaded areas in each figure by examining which combinations of sets A, B, and C overlap:

Figure (i): The shaded region is the area where *all three* sets overlap simultaneously. This is the triple intersection.

Figure (ii): The shaded region covers every area where *at least two* of the three sets overlap. This includes the pairwise intersections $A \cap B$, $B \cap C$, and $A \cap C$ (and their common triple intersection).

Figure (iii): Two disjoint shaded regions appear: the part of B that lies outside both A and C, and the part of $A \cap C$ that lies outside B.

Results

(i) $A \cap B \cap C$

(ii) $(A \cap B) \cup (B \cap C) \cup (A \cap C)$

(iii) $(B \cap A^c \cap C^c) \cup (A \cap C \cap B^c)$